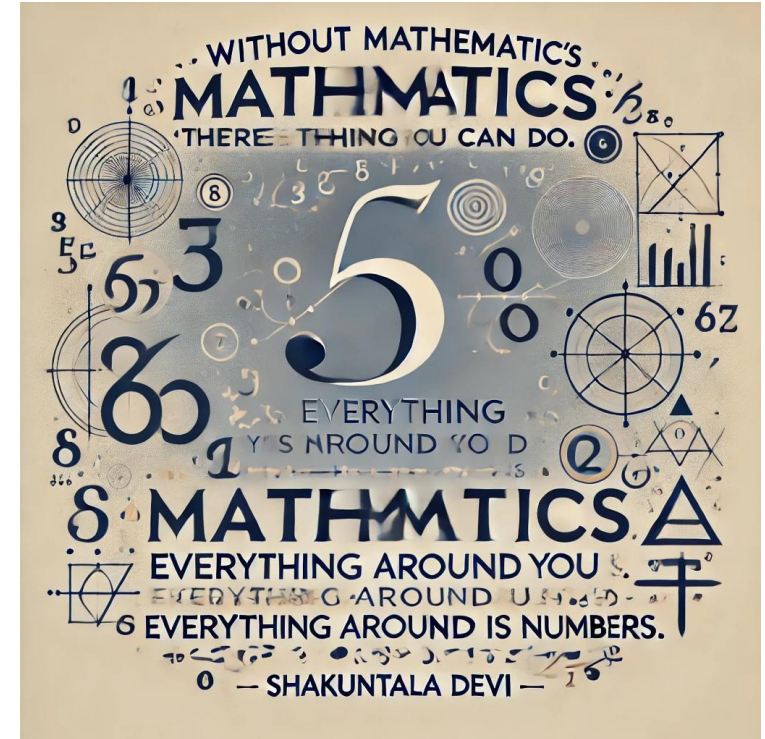


Mathematical Foundations of AI

The “thing” behind the “things” we are trying use to do “things”

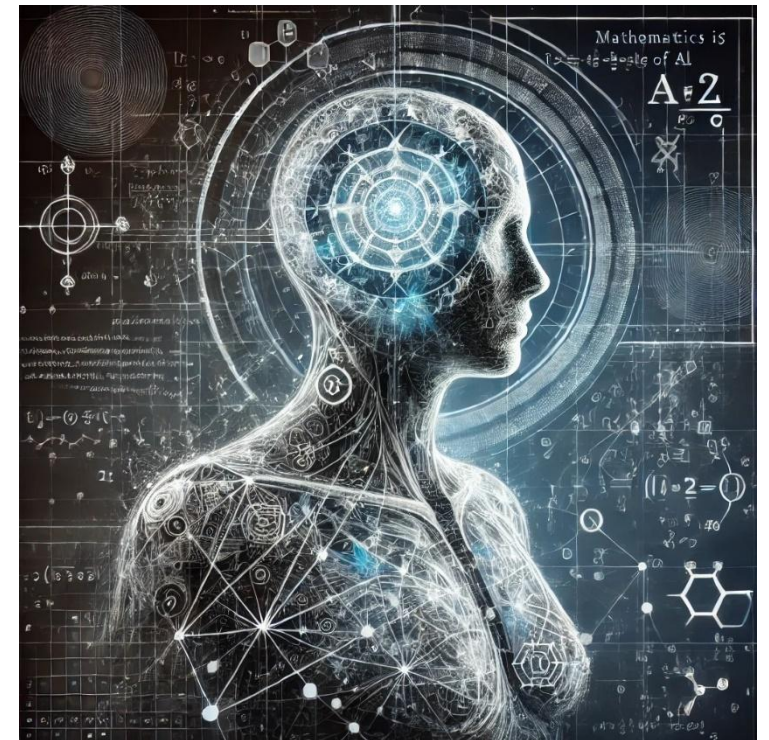
Introduction

- Simulation of human intelligence in machines
- Goal is to create systems capable of performing tasks:
 - recognizing speech, making decisions, or solving complex problems.



Importance of Mathematics in AI Development

- Mathematics is the backbone of AI
 - Model Formulation
 - Optimization
 - Probability and Statistics
 - Data Representation
 - Algorithm Design
 - Validation and Testing



Neural Network and Math

$$f(x_1, \dots, x_n) = \rho \left(\sum_{i=1}^n x_i w_i - b \right) = \rho((x, w) - b)$$

$$T_\ell : \mathbb{R}^{N_{\ell-1}} \rightarrow \mathbb{R}^{N_\ell}, \quad T_\ell x = W^{(\ell)} x + b^{(\ell)}, \quad W^{(\ell)} \in \mathbb{R}^{N_\ell \times N_{\ell-1}}$$

$$b^{(\ell)} \in \mathbb{R}^{N_\ell}$$

$$\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{N_L}$$

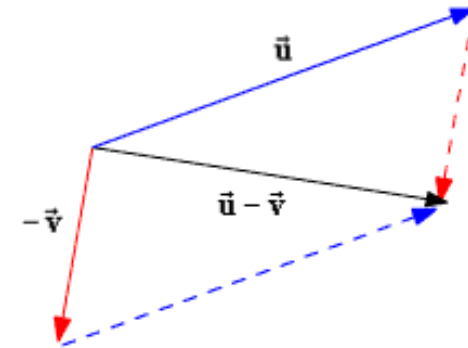
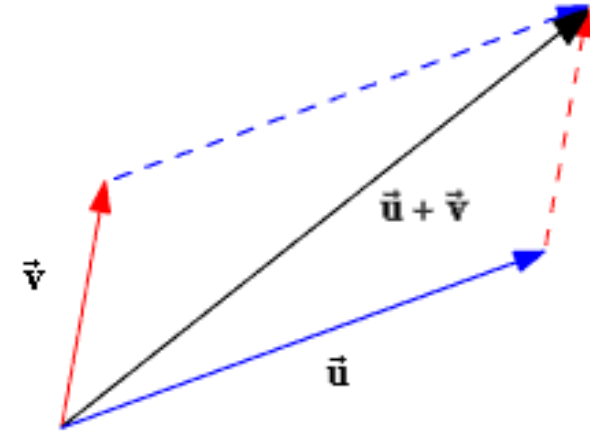
$$\Phi(x) = T_L \rho(T_{L-1} \rho(\dots \rho(T_1(x))))), \quad x \in \mathbb{R}^d$$

Linear Algebra

- Branch of mathematics concerning linear equations, linear functions and their rep.
- Data representation, transformation, optimization
- Scalars, Vectors and Matrices

Vector operations

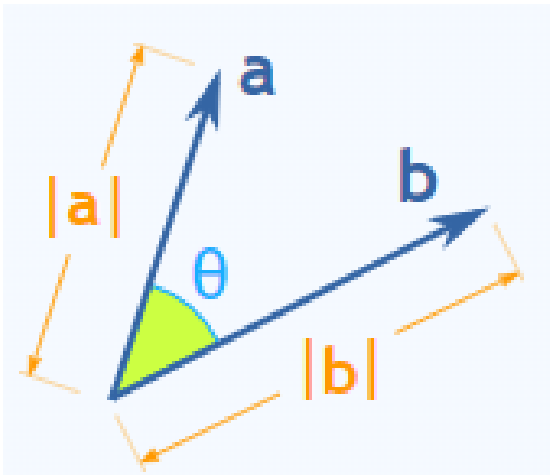
- Vector Addition and Subtraction
- Given two vectors $\vec{u}$ and $\vec{v}$, where $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ then
 - $\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2)$
 - $\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$
 $= (u_1 - v_1, u_2 - v_2)$



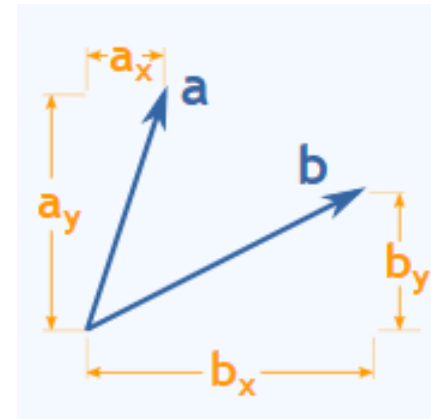
https://www.varsitytutors.com/hotmath_help/topics/adding-and-subtracting-vectors

Vector Operations: Dot product

- Often called inner product*
- Very important in AI



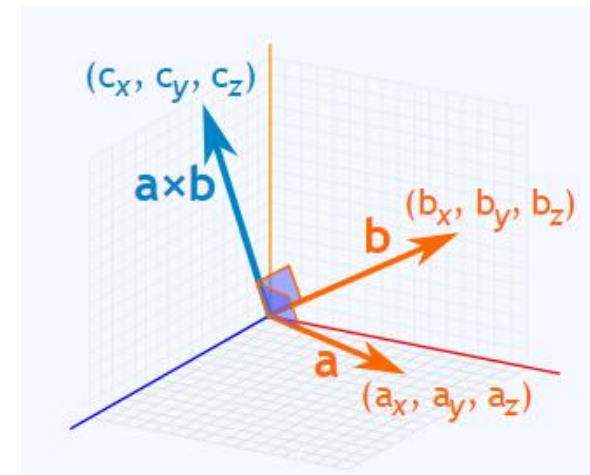
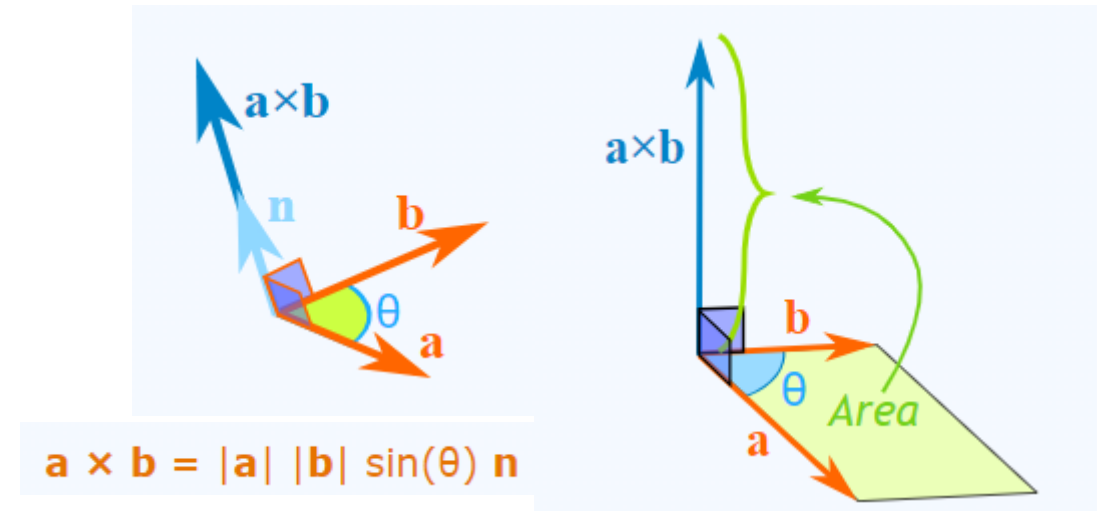
$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \times |\mathbf{b}| \times \cos(\theta)$$



$$\mathbf{a} \cdot \mathbf{b} = a_x \times b_x + a_y \times b_y$$

Vector Operations: cross product

- **another vector** that is at right angles to the original vectors
- Magnitude equals the area of a parallelogram with these two vectors for sides
- Important in computer graphics



$$\begin{aligned}c_x &= a_y b_z - a_z b_y \\c_y &= a_z b_x - a_x b_z \\c_z &= a_x b_y - a_y b_x\end{aligned}$$

Matrix Operations

- Matrix addition and multiplication
- Transpose of a Matrix
- Inverse of a Matrix
- Determinants and rank

Matrix Addition

$$\begin{bmatrix} 5 & 4 \\ 8 & 6 \\ 1 & 3 \end{bmatrix} + \begin{bmatrix} 9 & 2 \\ 8 & 6 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 5+9 & 4+2 \\ 8+8 & 6+6 \\ 1+3 & 3+0 \end{bmatrix} = \begin{bmatrix} 14 & 6 \\ 16 & 12 \\ 4 & 3 \end{bmatrix}$$

Matrix Subtraction

$$\begin{bmatrix} 7 & 6 & 9 \\ 4 & 8 & 2 \end{bmatrix} - \begin{bmatrix} 6 & 9 & 1 \\ 2 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 7-6 & 6-9 & 9-1 \\ 4-2 & 8-5 & 2-7 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 8 \\ 2 & 3 & -5 \end{bmatrix}$$

Matrix Multiplication

$$\begin{bmatrix} 7 & 6 & 1 \\ 2 & 3 & 8 \end{bmatrix} * \begin{bmatrix} 4 & 9 \\ 1 & 7 \\ 3 & 8 \end{bmatrix} = \begin{bmatrix} 7*4+6*1+1*3 & 7*9+6*7+1*8 \\ 2*4+3*1+8*3 & 2*9+3*7+8*8 \end{bmatrix} = \begin{bmatrix} 37 & 113 \\ 35 & 103 \end{bmatrix}$$

Transpose Of A Matrix

$$\begin{bmatrix} 7 & 2 & 9 \\ 1 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 7 & 1 \\ 2 & 6 \\ 9 & 3 \end{bmatrix}$$

Neural Network and Math

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$$T_\ell : \mathbb{R}^{N_{\ell-1}} \rightarrow \mathbb{R}^{N_\ell}, \quad T_\ell x = W^{(\ell)} x + b^{(\ell)}, \quad W^{(\ell)} \in \mathbb{R}^{N_\ell \times N_{\ell-1}}$$

$$b^{(\ell)} \in \mathbb{R}^{N_\ell}$$

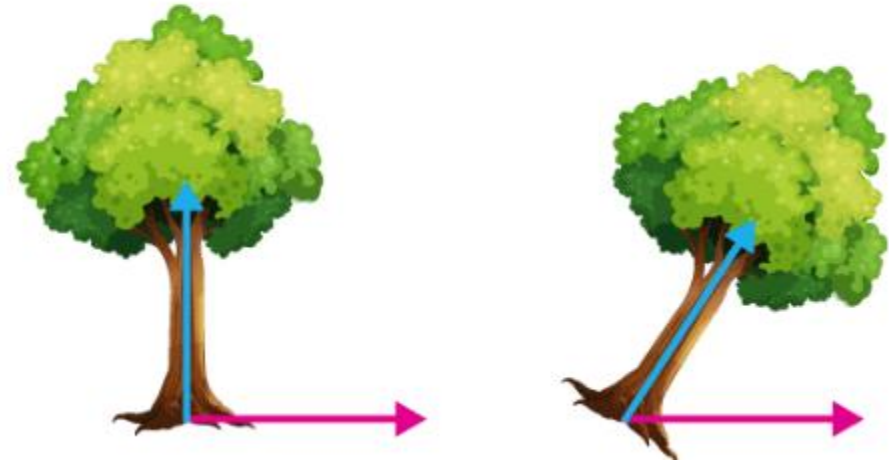
$$\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^{N_L}$$

$$\Phi(x) = T_L \rho(T_{L-1} \rho(\dots \rho(T_1(x))))), \quad x \in \mathbb{R}^d$$

Eigenvalue and Eigen vectors

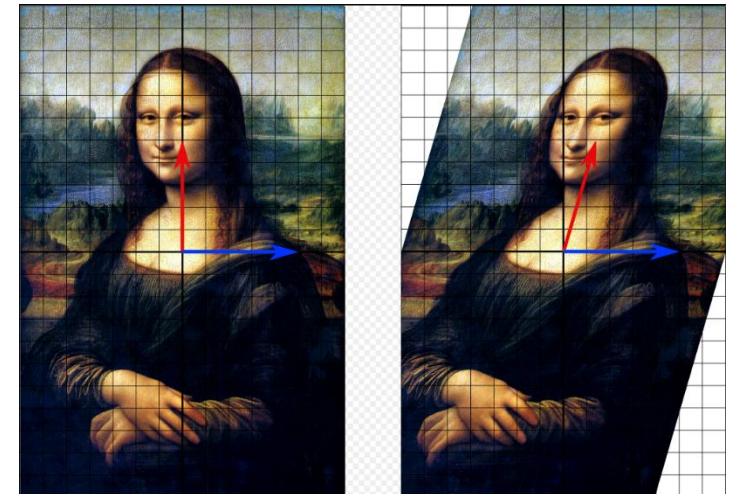
- Characteristic value, characteristic root, proper values or latent roots
- Eigenvectors changes by only a scalar factor

$$Ax = \lambda x$$



Eigen* and AI

- Understanding Linear Transformations
- Dimensionality Reduction: Principal Component Analysis (PCA)
- Spectral Clustering and Graph Analysis
- Optimization in Machine Learning Models
- Face Recognition and Image Compression



https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors

(more) Applications of Linear Algebra in AI

- Neural Networks: Weight matrices, input and output transformations.
- Natural Language Processing (NLP): Vector space models, word embeddings.
- Computer Vision: Image transformations, convolution operations.

Others

- Statistics and Probability
- Information theory: entropy, cross-entropy, Viterbi Algorithm
- Calculus

How can you learn

- Books
- Videos
- Tutorials
- Workshops
- Read
- Read again
- Then try and read

Questions?



Thank you
for listening



“Well, well, well...
look who’s trying to
end an argument
by giving me food.
If you think that’s
gonna work you’re
absolutely right”